

An Asymmetric Mechanism in a Symmetric Molecular Machine

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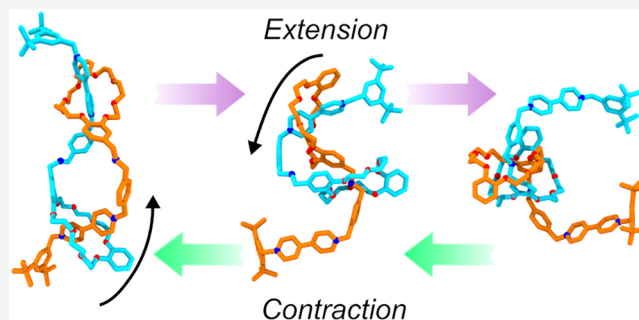


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Supporting Information

ABSTRACT: The design of molecular architectures exhibiting functional motions is a promising area for disruptive technological development. Toward this goal, rotaxanes and catenanes, which undergo relative motions of their subunits in response to external stimuli, are prime candidates. Here, we report on the computational analysis of the contraction/extension of a bistable [c2]daisy chain rotaxane. Using free-energy calculations and transition path optimizations, we explore the free-energy landscape governing the functional motions of a prototypical molecular machine with atomic resolution. The calculations reveal a sequential mechanism in which the asynchronous gliding of each ring is preferred over the concerted movement. Analysis of the underlying free-energy surface indicates that the formation of partially rearranged intermediates entails crossing of much smaller barriers. Our findings illustrate an important design principle for molecular machines, namely that efficient exploitation of thermal fluctuations may be realized by breaking down the large-scale functional motions into smaller steps.



The successful design of artificial molecular machines requires ways of eliciting and controlling relative motions at the molecular scale, similar to what is realized in complex biomolecules such as ATP synthase,¹ myosin,² or ligand-gated ion channels.³ Catenanes and rotaxanes are among the most promising synthetic architectures toward this goal.⁴ In bistable rotaxanes, a ring is threaded along a linear axle and prevented from escaping by terminal stoppers. Motion of the ring is controlled by the modulation of the differential binding affinity for topographically distinct recognition sites along the axle, referred to as *stations*, via external stimuli, e.g. protonation/deprotonation, oxidation/reduction, solvent change, temperature change, etc (see ref 5 for a review). A particular rotaxane-based motif, the [c2]daisy chain, makes use of a covalent link between ring and axle so as to obtain an interlocked architecture.^{6,7} When based on bistable rotaxanes, the resulting supra-molecule can undergo substantial extension/contraction motions very much like a muscle.⁸ Since its introduction, many variations on this theme have been explored^{9–11} with the ultimate goal of producing stimuli-responsive contractile materials.^{12–16} However, both thermodynamic and kinetic aspects underlying the extension/contraction motions, which are critical for the design of efficient devices, remain to be elucidated.

While inspiration from macroscopic devices has certainly been fruitful for the design of molecular machines, one should not forget that nanomachines operate in an environment dominated by viscous dissipation and thermal fluctuations.¹⁷ In this context, computational analyses based on all-atom molecular dynamics, which are time-resolved and include

finite-temperature effects, are instrumental to reveal otherwise hidden features. For instance, a recent simulation analysis unveiled an unexpected rotational motion of the ring upon gliding of a rotaxane along its axle.¹⁸ In addition, another study highlighted the role of the solvent on the modulation of the shuttling mechanism and its kinetics in amide-based rotaxanes.¹⁹ Therefore, the question as to whether synthetic chemists could/should better take into account the peculiarities of the nanoscale and seek inspiration from existing biological motors has been raised.²⁰

Here, using advanced molecular simulation techniques, i.e., free-energy calculations based on the Adaptive Biasing Force (ABF) method^{21–23} and path optimizations via the string method in collective variables,^{24,25} we explore the contraction/extension of a symmetric, acid–base controlled [c2]daisy chain rotaxane previously reported by Stoddart and co-workers¹¹ (see Figure 1). The simulations indicate that the most likely rearrangement proceeds through an asymmetric mechanism, which involves gliding of the crown-ether rings along the rotaxane axle one after the other. Using free-energy calculations, we show that such an asynchronous mechanism, which is enabled by the intrinsic flexibility of the axle, reduces the shuttling barrier by a factor of 2, thus allowing for a more

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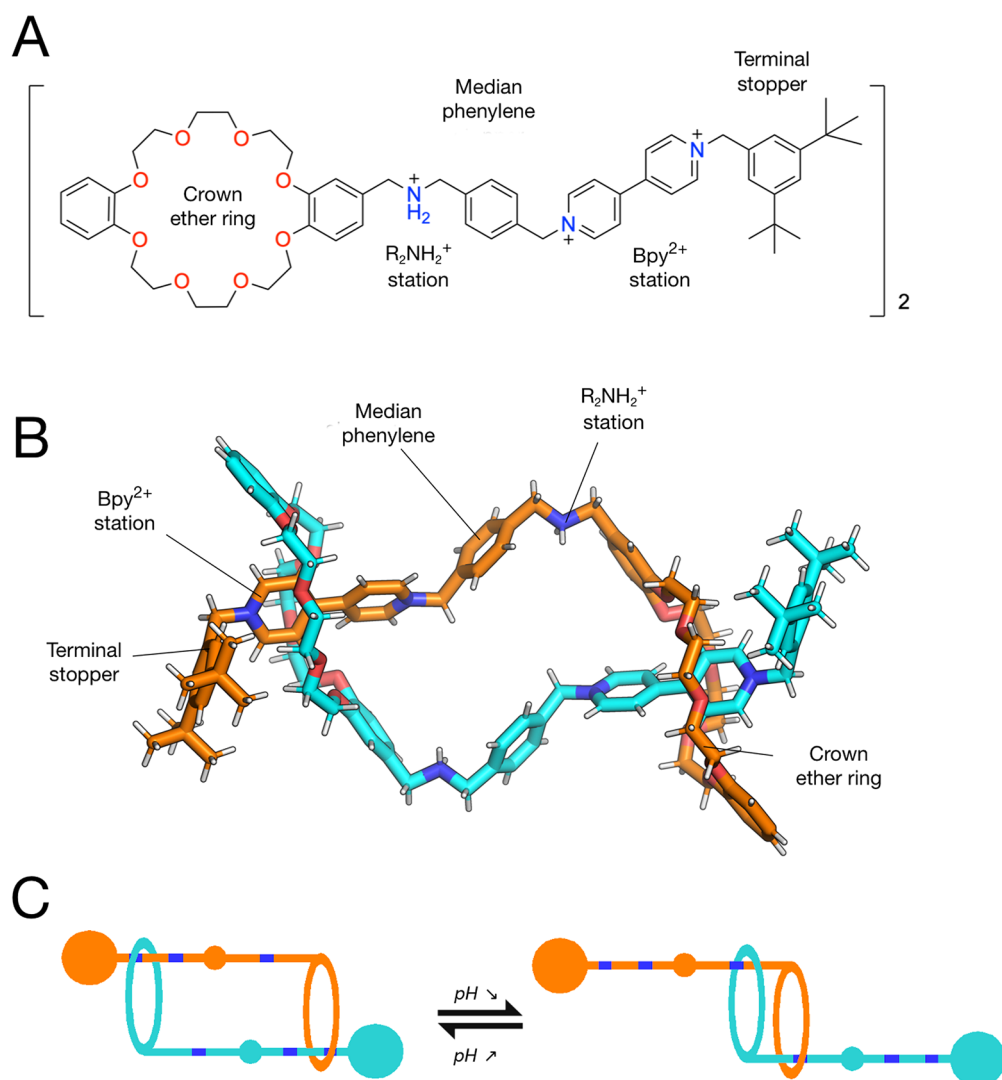


Figure 1. Overview of the system and schematic representation of the gliding mechanism as anticipated from NMR studies. (A) Chemical structure of the [c2]daisy chain rotaxane. Significant sites are indicated. (B) Molecular model of the [c2]daisy chain in the contracted form. Significant sites are highlighted on the orange subunit only. (C) Schematic representation of the contracted and extended states. Stations are highlighted by blue stretches. The scheme illustrates how the crown-ether rings must cross the median phenylene groups, represented here as smaller bullet points, during isomerization.

efficient interconversion between the contracted and extended forms. The present findings illustrate in the clearest fashion an important design principle for molecular machines.

To explore the daisy-chain rotaxane by molecular dynamics, we built an atomistic model of the complex with the CHARMM/CGENFF^{26–30} energy function and equilibrated it at 300 K and 1 bar in a simulation box filled with acetonitrile molecules (see the Supporting Information for details on modeling). The system was modeled with a protonated $R_2NH_2^+$ station, which corresponds to low-pH conditions. As evidenced by NMR, in these conditions the crown-ether groups preferably bind to this recognition site and the 4,4'-bipyridinium (Bpy^{2+}) station is essentially empty.¹¹ We stress that pH variations are not the focus of this study; rather, we are interested in the energetics and the molecular mechanism of bistable switching that takes place when both recognition sites along the axle have significant affinity for the ring, i.e., at a generic low pH. Four collective variables (CVs) were introduced to characterize the functional motions of the daisy-chain rotaxane. We defined P_i ($i = 1, 2$) as the position of

ring i along its axle¹⁸ (note that P_1 and P_2 are symmetric by definition), the total extension L as the end-to-end distance, and the inner extension l as the distance between the two rings (see Figure S2 for a schematic representation of the CVs). P_1 and P_2 were used as supporting CVs to run the free-energy calculations with the ABF strategy, which allows for efficient mapping of the free-energy landscape of (relative) ring gliding. In addition, all four CVs were used in a string method optimization to obtain the transition path of minimum free energy from contracted to extended (see the Supporting Information for details on the calculations).

The free-energy landscape along P_1 and P_2 computed by ABF and symmetrized by arithmetic averaging is shown in Figure 2. Symmetrization was needed to correct for incomplete convergence (a fully converged free-energy landscape would be perfectly symmetric on account of the symmetry of the CV definitions), but even without it the landscape exhibits qualitative symmetry (see Figure S3). Free-energy basins corresponding to the contracted and extended forms are located along the main diagonal, with the extended state being

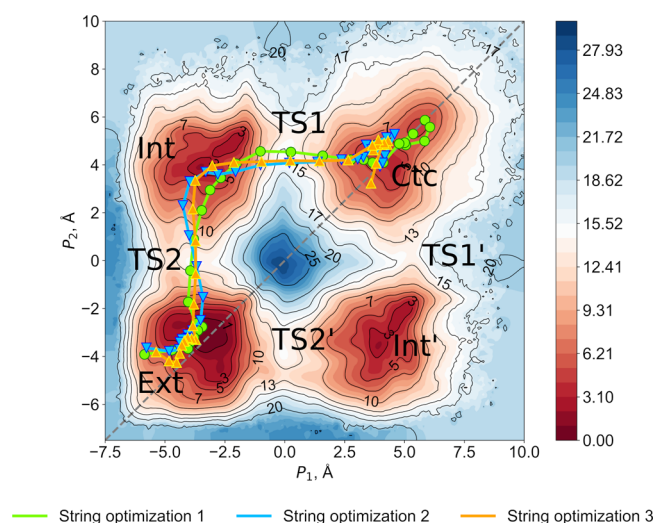


Figure 2. Free-energy landscape of contraction/extension of the [c2]daisy chain computed by ABF calculations. The free energy is expressed in kilocalories per mole and has been symmetrized with respect to the main diagonal. Also shown are the projections of the three independent optimized strings onto the (P_1, P_2) -space. Remarkable states are labeled as follows: Ctc, contracted state; TS1, transition state 1; Int, off-diagonal intermediate; TS2, transition state 2; Ext, extended state. Symmetry-equivalent states on the lower-right portion of the plot are denoted with the prime symbol ($'$). The iso-contour lines indicate that the free-energy barrier for asymmetric gliding through an off-diagonal intermediate is about half of that for synchronous contraction/extension along the diagonal.

the global minimum in agreement with NMR experiments.¹¹ Perhaps surprisingly, off-diagonal metastable states are also found. Based on this map, two symmetry-equivalent and low free-energy pathways connecting the extended and contracted states can be identified. These paths bypass the main diagonal and proceed via the off-diagonal intermediates. Thus, the calculation predicts a sequential, asymmetric transition for the contraction/extension of this daisy-chain rotaxane. Also, it indicates that a fully symmetric mechanism would entail a free-energy barrier on the order of 24 kcal mol^{-1} , which suggests that asynchronous gliding would be about one hundred million times faster than the concerted symmetric mechanism (see the [Supporting Information](#) for an estimation of the acceleration factor by asymmetric gliding).

Path optimizations with the string method in the subspace defined by P_1 , P_2 , L , and l were used to determine the most likely transition pathway independently and to obtain a structural representation of it. The string method results robustly confirm the prediction from ABF (Figure 2) and open to the structural characterization of the off-diagonal intermediates with atomic resolution. Moreover, umbrella sampling along the optimized string shows semiquantitative agreement of the resulting free-energy profile with the ABF results (Figures 3 and S12). Interestingly, the umbrella sampling calculation reveals that the low-energy intermediates correspond to half-glided configurations of the daisy chain rotaxane in which only one ring has passed the median phenylene group (see Figure 3). In these configurations, significant bending (“buckling”) of the axle is observed, demonstrating the importance of axle flexing in the asymmetric mechanism. Altogether, these results suggest that asymmetric, sequential (off-diagonal) gliding is preferred over the symmetric, concerted (on-diagonal) mechanism because it

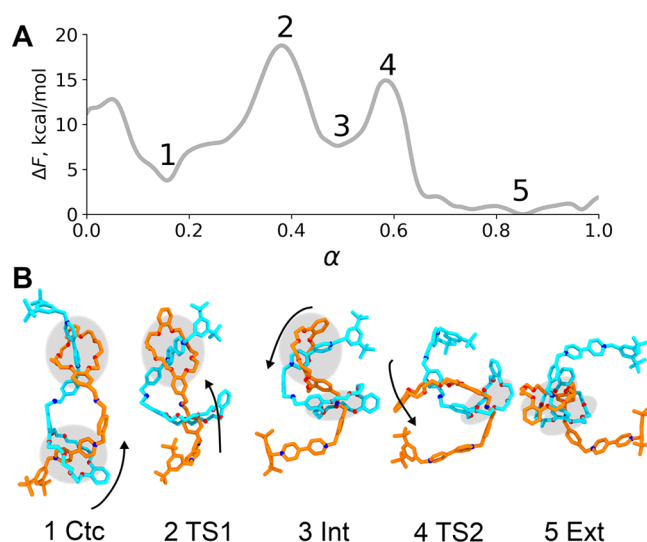


Figure 3. (A) Free-energy profile along the most probable transition path. Shown is the free-energy profile computed along string 1 by on-the-path umbrella sampling in the four-dimensional space. (B) Structural mechanism of the transition. Representative conformers of the remarkable states extracted from the string calculation are shown. Black arrows represent the direction of the crown ether ring movements. Shaded gray ellipses indicate stabilizing interactions between the rings and the stations. In the contracted form (Ctc), both rings interact with their respective Bpy^{2+} stations providing stability to this conformation. The transition is initiated by an inward gliding movement of the ring of the cyan subunit (conventionally) which breaks its interaction with the corresponding Bpy^{2+} site. The first transition state (TS1) corresponds to a configuration in which the cyan ring is no longer stabilized by interaction with Bpy^{2+} and not yet by interaction with R_2NH_2^+ . After the cyan ring completes its gliding, it interacts with R_2NH_2^+ while the orange ring is still stabilized by its Bpy^{2+} station; the corresponding configuration is thus a metastable intermediate (Int). In the second stage of the transition, the orange ring breaks its interactions with its Bpy^{2+} station, visits a high-energy configuration (TS2) in which it does not interact with any station, and finally completes gliding by binding to its R_2NH_2^+ station. The resulting configuration has both rings interacting with their respective R_2NH_2^+ stations, providing global stability to this extended state (Ext). It is clear from the figures that one-sided ring gliding entails significant bending of the associated axle, highlighting how molecular flexibility is critical to explore the asymmetric pathway.

decomposes a conformational transition involving one large free-energy barrier into two subtransitions corresponding to energetically less costly elementary steps. Also, they highlight that axle flexibility plays a role in the functional isomerization as it provides a lower-energy transition path relative to symmetric gliding. The existence of half-glided, asymmetric intermediates in this pH-responsive daisy chain could be, in principle, verified by NMR. Recently, a joint experimental/computational study of an electrochemically controlled bistable daisy-chain rotaxane suggested the existence of half-glided intermediates.⁹

In order to provide a chemical understanding of the asymmetric mechanism highlighted by the free-energy calculations, an energy component analysis was carried out along the minimum free-energy path provided by the string method. The results of the analysis show that the main energetic component to the barriers along the path is of electrostatic nature and corresponds to the loss of hydrogen bonds between the gliding crown-ether ring and the two

stations rather than steric hindrance by the median phenylene group (Figure S11). Thus, concerted gliding appears to be kinetically disfavored because it entails the simultaneous breaking of twice as many hydrogen bonds relative to the asymmetric mechanism. Along this line, by lowering the energy cost of H-bond breaking, a more polar solvent such as water is expected to make the transition faster, consistent with recent results by Fu and co-workers.¹⁹ Whether the stabilizing effect of water would be strong enough to lower or even reverse the kinetic advantage of the asymmetric mechanism via axle bending is currently unknown.

Visual inspection of the string optimization results indicate that in the extended state the flexible tails may adopt a folded conformation with the end-to-end distance being as short as 10 Å (see Figure 3B). To quantify the relative abundance of folded conformers at equilibrium, an additional 600 ns of unbiased MD simulation was run starting from an extended configuration of the daisy chain. The effective free-energy profile obtained by Boltzmann-inversion of the statistical distribution of L is shown as Figure S1. The profile exhibits two minima. The global minimum at $L = 35$ Å corresponds to configurations of the daisy chain with elongated tails, while the secondary minimum at $L = 14$ Å that is about 0.75 kcal mol⁻¹ higher in free energy includes more compact configurations of the tails similar to those captured by the string calculation. The flatness of the free-energy surface along L (~ 1 kcal mol⁻¹ total free-energy drop for L in the 10–40 Å range) indicates fast and reversible interconversion between the compact and elongated configurations of the tails, which highlights the intrinsic flexibility of the extended form of this daisy chain in solution. As expected, the bistable behavior of the tails disappears in the contracted state (data not shown).

Despite possible inaccuracies inherent to the modeling, the numerical findings above bear significant implications for molecular design. First, our analysis highlights how (chemical) intuition, which is often based on the analogy with macroscopic objects, may yield an inaccurate or incomplete picture of molecular phenomena. Here, the striking observation is that sequentiality of motions rather than concertation allows for a much faster isomerization of the [c2]daisy chain because large free-energy barriers are broken down into energetically less costly substeps. Because the exploration of half-glided intermediates would be impossible with perfectly rigid axles, this finding highlights that conformational compliance in the axles promotes faster isomerization of the daisy chain via asymmetric gliding, which is expected to increase both the sensitivity and the speed of response to external stimuli. As such, the present simulations bring new perspectives to the long-standing debate of flexibility vs rigidity in the assembly of complex supramolecular architectures³¹ and suggests that strong coupling between rigid molecular components, which is sought after for efficient force transmission in macroscopic devices, is not a requirement for the operation of molecular machines at the nanoscale. On the contrary, axle flexibility emerges as a tunable molecular parameter to achieve fine control over the functional movements, along with external parameters such as solvent, pH, and ionic strength.³² This design strategy could be applied to improve existing functional architectures. For example, Sauvage and co-workers reported that a transition-metal-based bistable rotaxane failed to exhibit contraction on an observable time scale after oxidation of copper(I) to copper(II), which was attributed to the requirement that both rings should move in concert.¹³ If so,

increasing axle flexibility should alleviate this problem. More generally, by relieving the need for simultaneous gliding, our strategy should prove useful to improve actuation of more complex rotaxane-based architectures, such as molecular presses and elevators.¹² We note on passing that although the existence of stepwise molecular motions underlying the operation of multicomponent catenanes and rotaxanes has been previously reported,^{33,34} their asymmetric nature stems from chemical dissymmetrization of the supramolecule upon application of an external stimulus to drive the isomerization; for example, in the work of Meng et al.³⁴ the stepwise migration of the crown ether rings was promoted by gradual introduction of a base that results in the sequential deprotonation of the dibenzylammonium stations as evidenced by distinct pK_a values per station. In sharp contrast, the sequential mechanism described here follows from the inherent properties of the underlying free-energy landscape (i.e., pathways and barriers) and is therefore of intrinsic origin.

Second, the present calculations highlight the existence of compact molecular structures even in the extended state of the daisy chain with the flexible tails adopting a folded conformation possibly stabilized by tail–tail interactions. Although the existence of such conformers is not surprising *per se* (i.e., a fully extended conformation is expected to suffer from some entropic penalty in a flexible molecule) and was previously proposed based on DFT calculations,³⁵ the latter would translate molecular extension into an even shorter end-to-end distance, thus challenging the applicability of this architecture in the context of molecular muscles. Our free-energy analysis of the extended-state basin, however, reveals that the folded and the unfolded configurations of the tails are in reversible equilibrium, such that folding of the tails is not detrimental for function, particularly in the presence of external pulling forces that would oppose folding. Whether external forces would also affect the mechanism and/or the kinetics of gliding remains an open question that will be addressed elsewhere.

Last, although rotaxane architectures are far less complex than functional biomolecules such as enzymes, motors, and switches, the supramolecule investigated here is suggestive and may be used as a model system to explore the fundamental principles of biomolecular machine operation. Under this hypothesis, our findings suggest that an efficient exploitation of thermal fluctuations by molecular machines may be realized by breaking down the large-scale functional motions into smaller substeps. In addition, they indicate that symmetry (whenever applicable) should not be necessarily preserved during a functional isomerization. If these conclusions were general, the existence of structural intermediates corresponding to partially rearranged metastable states should be common in functional nanomachines in both chemistry and biology. Consistent with this hypothesis, Okazaki and Hummer have come to the conclusion that three rotational substeps (rather than two) along the functional cycle of mammalian ATP synthase are required to avoid too high free-energy barriers.³⁶ In addition, recent cryo-EM studies of *Polytomella* sp. mitochondrial ATP synthase have demonstrated the existence of several intermediates due to weak coupling between the rotary and static portions of this motor.³⁷ And, a weak-coupling mechanism along with the corresponding intermediate has been characterized along the recovery stroke of the myosin biological motor.³⁸ Whether sequential asymmetric transitions must be a hallmark even in symmetric multimeric architectures

(such as hemoglobin or the chaperonine GroEL) is presently unclear and requires further investigation. We note, however, that microsecond simulations of the 5-fold symmetric proton-gated ion channel GLIC suggested a sequential, asymmetric mechanism for pore closing.³⁹ Although these conclusions seem to challenge a central tenet of the celebrated Monod–Wyman–Changeux (MWC) model of allostery,⁴⁰ this is in fact not the case as this model assumes symmetry only at the initial and final states, a condition that is satisfied by the synthetic machine investigated here.

In conclusion, the rational design of functional architectures is currently driven by chemical intuition and may overlook important aspects in the absence of time-resolved and/or atomically detailed analyses. We have analyzed the gliding mechanism for contraction/extension in a bistable, doubly interlocked, daisy-chain rotaxane by molecular dynamics and advanced free-energy calculations. Together with recent simulation analyses of analogous molecular machines, our study illustrates how all-atom simulations may provide access to the design principles of functional supramolecular architectures.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcllett.1c00404>.

Detailed materials and methods, estimation of the acceleration factor, supplementary figures, and supplementary references (PDF)

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Notes

The authors declare no competing financial interest.

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